# LESSON PLAN

## BASIC ELECTRICAL (23)

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| **Class** | Air Agniveer Apprentice (Engineering) | **Lesson Plan No.** | 15 |
| **Total No. of Lesson Plans** | 29 | **Time Allotted for the Session** | 80 min |

## PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

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| **1. Topic** | Tracing and identification of various under carriage components, Shock observers, types, brief description and operation |
| **2. Total No. of Sessions for this Topic** | 01 / 01 |
| **3. Sequence of this Session** | 01 / 01 |
| **4. Syllabus Extract** | Tracing and identification of various under carriage components, Shock observers, types, brief description and operation |
| **5. Specific Objectives** | 23.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction. |

**6. List of Training Aids**

| Ser | Description | Remarks |
|---|---|---|
| (a) | BR/Text Book/Docket | Refer to the master syllabus reference list |
| (b) | Wall Charts | Nil |
| (c) | Films/Duration | Nil |
| (d) | PPT | Objective: teach the syllabus extract for this lesson |
| (e) | Models | Nil |
| (f) | CBT | Nil |

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| **7. Practical in Lab/Equipment Rooms/Simulators** | Identification of under carriage components and shock absorber types on aircraft training equipment; Safety: Ensure training equipment stability and no loose parts; Supervision: Instructor monitors trainee handling of components. |
| **8. Existing Knowledge of the Class** | As per course prerequisites |

## PART B: DEVELOPMENT OF LESSON

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| **9. Checking of Previous Knowledge / Introduction** | Time: 10 min |

(a) What are the primary functions of the aircraft landing gear?
(b) Name the three main types of landing gear configurations.
(c) Define the term 'shock absorber' in the context of landing gear.

**10. Development of Main Lesson**

| Sub | Content | Method | Trg Aids | Time |
|---|---|---|---|---|
| (a) | • Landing gear assembly overview<br>• Main and nose gear identification<br>• Wheel well and strut components<br>• Hydraulic lines and actuators<br>• Brake assembly location | L & Ds | PPT, IB | 15 min |
| (b) | • Shock absorber types overview<br>• Oil-air shock absorber operation<br>• Nylon cord shock absorber operation<br>• Mechanical spring shock absorber<br>• Telescopic strut construction | L & Ds | PPT, PDF | 20 min |
| (c) | • Shock absorber maintenance basics<br>• Common failure modes identification<br>• Operational checks and testing<br>• Component replacement procedures<br>• Safety precautions during inspection | Practical | IB, PDF | 15 min |

**11. Summary and Evaluation**

| Method | Trg Aids | Time |
|---|---|---|
| Q & A | IB | 15 min |

(a) Explain the difference between oil-air and nylon cord shock absorbers.
(b) Describe the construction of a typical telescopic strut.
(c) List the key components found in the main landing gear assembly.

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| **12. Home Assignment and Follow Up** | Prepare a brief note on the differences between main wheel and nose wheel configurations for the next lesson (LP 16). |
| **Time** | 5 min |

## PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

| Ser | Syllabus for PFT | References | Related LPs |
|---|---|---|---|
| 1 | Wheels, types of wheels and description of typical main wheel | Refer to the master syllabus reference list | LP 16 |

## PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

| Ser | Item | Remarks |
|---|---|---|
| (a) | Adequacy of time | |
| (b) | Suitability of training material/aids and additional aids required | |
| (c) | Doubts/intelligent questions raised by trainees | |
| (d) | Difficulties faced by instructor | |
| (e) | Other experience | |

| **Instructor** | **HoF (E/L)** |
|---|---|
| Signature: _____ | Signature: _____ |

## PART E: REVIEW SHEET

| Date | Instructor's Comment | HOF (E/L) | Oi/C TDEC Remarks |
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**Prepared by / Approved by**

| Prepared by | Approved by |
|---|---|
| Name, Rank, P No. | Name, Rank, P No. |
| Nov 26 | Nov 26 |

## Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

## Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min